

AI-Driven Cyber Threat Intelligence Framework for Early Detection of Advanced Persistent Threats Using Behavioral and Tactical Mapping

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ABSTRACT- Advanced Persistent Threats (APTs) represent a critical challenge in modern cybersecurity due to their stealthy, multi-stage, and long-duration nature. Traditional intrusion detection systems and Security Information and Event Management (SIEM) platforms rely heavily on signature-based methods and rule-based correlation, rendering them ineffective against previously unseen, evasive attack patterns. This paper proposes a novel AI-driven Cyber Threat Intelligence (CTI) framework that integrates machine learning-based behavioral analysis with tactical mapping to the MITRE ATT&CK framework for early-stage APT detection. The framework comprises a multi-layer architecture incorporating network telemetry, endpoint logs, and open-source intelligence feeds. A hybrid AI detection engine combining XGBoost for feature classification and Long Short-Term Memory (LSTM) networks for sequential pattern learning identifies anomalous behaviors indicative of APT campaigns. Detected anomalies are subsequently mapped to MITRE ATT&CK techniques through a tactical mapping engine, enabling security analysts to understand adversary intent and progression within the cyber kill chain. The framework is validated using the UNSW-NB15 dataset and a custom APT simulation environment, achieving a detection accuracy of 96.8%, precision of 94.2%, and recall of 95.7%, with a false positive rate of 2.9%. Comparative analysis demonstrates significant improvements over traditional SIEM systems in detecting multi-stage attacks and reducing alert fatigue. The key contributions include a unified intelligence-driven detection architecture, a hybrid AI model optimized for APT behavioral patterns, and an automated tactical mapping mechanism that bridges the gap between raw telemetry and strategic threat intelligence

Keywords—Cyber Threat Intelligence, APT Detection, MITRE ATT&CK, Machine Learning, Behavioral Analysis, Predictive Security, XGBoost, LSTM.

I. INTRODUCTION

The landscape of cyber threats has undergone a profound transformation over the past decade. What once consisted

opportunistic malware campaigns and financially motivated cybercriminals has evolved into a sophisticated ecosystem of state-sponsored actors, organized cybercrime syndicates, and

hacktivist collectives employing highly targeted, stealthy, and persistent attack methodologies. This evolution has been accelerated by the rapid digitization of critical infrastructure, the proliferation of cloud services, and the expanding attack surface introduced by the Internet of Things (IoT) and remote work paradigms [1]. Among the most concerning developments in this threat landscape is the emergence and proliferation of Advanced Persistent Threats (APTs). APTs are characterized by well-resourced adversaries who conduct prolonged, multi-stage intrusion campaigns against specific targets, often remaining undetected within compromised networks for months or even years [2]. Unlike conventional cyberattacks that seek immediate financial gain or service disruption, APTs are typically motivated by strategic objectives including intellectual property theft, geopolitical intelligence gathering, critical infrastructure sabotage, and long-term network persistence for future operations. The defining characteristics of APTs include their stealthy nature, use of custom malware and zero-day exploits, extensive reconnaissance, lateral movement across networks, and the establishment of multiple redundant command-and-control channels. High-profile APT campaigns, such as those conducted by the Lazarus Group against financial institutions or the Sunburst supply chain attack, demonstrate the devastating potential of these threats to cause billions in economic damage and compromise national security interests [3]. In response to this evolving threat landscape, the cybersecurity community has recognized the imperative to shift from reactive, signature-based defense mechanisms toward proactive, intelligence-driven security postures. This paradigm shift emphasizes the continuous collection and analysis of threat intelligence, behavioral monitoring, and the adoption of adversary-centric defense frameworks that anticipate attacker movements rather than simply responding to successful breaches. The integration of threat intelligence into security operations enables organizations to prioritize defenses based on real-world

adversary tactics and to detect malicious activity earlier in the attack lifecycle.

A. Research Problem

Despite significant advances in cybersecurity technology, traditional detection systems continue to struggle against sophisticated APT campaigns. Several fundamental limitations undermine the effectiveness of conventional approaches: **First**, traditional Intrusion Detection Systems (IDS) and SIEM platforms predominantly rely on signature-based detection methodologies that compare observed events against databases of known attack patterns. While effective against commodity malware and well-documented exploits, signature-based approaches are inherently incapable of detecting novel, zero-day attacks or customized malware variants that do not match existing signatures. APT adversaries routinely employ custom tools and modify their techniques specifically to evade signature-based detection [4] **Second**, the multi-stage nature of APT campaigns poses significant challenges for detection systems that analyze events in isolation. APTs unfold over extended timeframes through a sequence of seemingly benign activities that, when considered individually, may not trigger alerts. A legitimate user credential compromised through spear-phishing may be used weeks later for lateral movement, followed by gradual data exfiltration spread across months. Traditional systems lack the temporal and contextual awareness necessary to correlate these distributed events into a coherent attack narrative

Third, the phenomenon of alert fatigue severely compromises the effectiveness of security operations centers (SOCs). Modern enterprises generate millions of security events daily, with SIEM systems producing thousands of alerts, the vast majority of which are false positives or low-severity events. This overwhelming volume desensitizes analysts to genuine threats and increases mean time to detect (MTTD) and respond (MTTR) [5]

B. Research Objectives

This research aims to address the identified gaps through the design, implementation, and validation of an integrated AI-driven Cyber Threat Intelligence framework for APT detection. The specific objectives are: -To design and develop a multi-source data collection and fusion architecture that integrates network telemetry, endpoint system logs, and open-source threat intelligence feeds into a unified analytical pipeline.

-To implement a hybrid AI detection engine combining gradient boosting (XGBoost) for efficient feature-based classification with Long Short-Term Memory (LSTM) networks for learning sequential behavioral patterns characteristic of multi-stage APT campaigns

To develop a tactical mapping engine that automatically correlates detected anomalous behaviors with MITRE ATT&CK tactics and techniques, providing security analysts with contextual understanding of adversary actions and campaign progression.

C. Research Questions

-How can multi-source telemetry (network, endpoint, and threat intelligence) be effectively fused to provide comprehensive visibility into APT campaign activities? - What hybrid machine learning architecture optimally balances detection accuracy and computational efficiency for real-time APT detection? -To what extent can detected behavioral anomalies be accurately mapped to specific MITRE ATT&CK techniques, and how does this tactical mapping enhance analyst understanding of attack progression?

-What are the comparative detection performance metrics (accuracy, precision, recall, false positive rate) of the proposed framework versus traditional signature-based systems and standalone ML models?

II. Ease of Use

The APTIntel-Sync framework is designed with a strong emphasis on usability and practical deployment in operational security environments. This section outlines the key features that enhance accessibility for security practitioners and the framework's applicability across various enterprise use cases

A. User-Centric Dashboard Design

The framework incorporates a web-based dashboard developed using React and D3.js visualization libraries, providing security analysts with intuitive access to detection results, tactical mappings, and investigation capabilities [6]. The dashboard is organized around the operational workflow of threat hunting and incident response:

Alert Prioritization Interface: Incoming alerts are displayed in a dynamically updated queue with color-coded severity indicators derived from the risk scoring model. Analysts can filter alerts by severity, MITRE tactic, affected assets, or time windows, enabling focused attention on the most critical threats. The interface presents key contextual information for each alert, including the confidence score from the AI detection engine, the mapped MITRE techniques, and the calculated risk score [7]

Attack Flow Visualization: For detected multi-stage attacks, the dashboard renders interactive attack flow diagrams based on MITRE Attack Flow v3.0.0 specifications [28]. Analysts can visualize the sequence of observed techniques, causal dependencies between stages, and parallel attack paths,

transforming abstract alert data into an intuitive narrative of adversary activity [8].

B. Simplified Investigation Workflow

The framework streamlines the investigation process through integrated workbench capabilities:

One-Click Telemetry Access: From any alert, analysts can instantly retrieve all raw telemetry associated with the detection, including network flows, process logs, and authentication events. The system automatically correlates related events across time and assets, presenting a unified timeline of suspicious activity [9]

Guided Investigation Prompts: Based on the mapped MITRE techniques, the system suggests investigation questions and next steps. For example, if credential dumping is detected, the interface prompts analysts to check for subsequent lateral movement attempts and provides pre-built queries for common post-exploitation indicators [10].

C. Practical Applications in Enterprise Environments

The framework addresses several practical use cases in enterprise security operations:

Continuous APT Monitoring: Organizations can deploy the framework for 24/7 monitoring of critical assets, providing early warning of sophisticated threat activity. The low false positive rate (2.9%) ensures that analysts can trust and act on alerts without constant alert fatigue.

Threat Hunting Support: Threat hunters use the framework's tactical mapping and attack flow reconstructions to identify indicators of compromise that may have been missed by automated detection. The visualization of adversary techniques guides hypothesis-driven hunting

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Compliance and Reporting: The framework generates compliance reports mapping detected threats to regulatory requirements (GDPR, HIPAA, PCI-DSS) and industry standards (NIST, ISO 27001), supporting audit and governance functions.

III. RELATED WORK

The development of integrated frameworks for APT detection using cyber threat intelligence and machine learning builds upon a substantial body of prior research spanning multiple

disciplines. This section reviews relevant literature across four key areas: machine learning in cybersecurity, behavioral anomaly detection, threat intelligence platforms and MITRE ATT&CK integration, and existing AI-driven APT detection frameworks

A. Machine Learning in Cybersecurity

The application of machine learning to cybersecurity has evolved substantially over the past decade. Early work focused on supervised learning approaches using Support Vector Machines (SVM) and decision trees applied to network flow features extracted from benchmark datasets such as KDD Cup 99 and NSL-KDD [12]. Tavallaee et al. [13] conducted comprehensive

evaluations of these approaches, demonstrating detection accuracies of 89-92% but noting significant limitations due to dataset age and the absence of contemporary attack patterns

More recent research has employed ensemble methods, particularly Random Forest and XGBoost, for intrusion detection tasks. Zhang et al. [14] applied XGBoost to the UNSW-NB15 dataset, achieving 94.3% accuracy and 91.7% recall for multi-class attack classification. Similarly, Kumar et al. [15] compared gradient boosting approaches with deep learning on the CIC-IDS-2017 dataset, finding that XGBoost achieved comparable accuracy to neural networks with significantly lower training and inference times

Deep learning approaches have shown particular promise for capturing complex patterns in security data. Vinayakumar et al. [16] proposed a deep neural network architecture for intrusion detection, evaluating multiple configurations on benchmark datasets. Convolutional neural networks (CNNs) have been applied to network traffic represented as images, with Wang et al. [17] achieving 94.5% accuracy by transforming packet payloads into grayscale images

B. Threat Intelligence Platforms and MITRE ATT&CK

The integration of threat intelligence into security operations has accelerated with widespread adoption of the MITRE ATT&CK framework [18]. ATT&CK provides a comprehensive taxonomy of adversary tactics, techniques, and procedures (TTPs) based on real-world observations, enabling standardized communication about threat behaviors [19]. Recent research has explored automated mapping of threat intelligence to ATT&CK techniques. Alshehri et al. [20] developed OTuHunt, a framework employing Natural Language Processing to extract IoCs and TTPs from unstructured CTI reports, achieving 87% accuracy in technique classification. The MITRE Center for Threat-Informed Defense developed TRAM (Threat Report Attack Mapper), which

automates mapping of CTI reports to ATT&CK with 82% accuracy [21]. Attack Flow v3.0.0 extends ATT&CK capabilities by providing a structured language for describing sequences of adversary behaviors, enabling visualization of parallel attack paths and logical dependencies [22]

C. AI-Driven APT Detection Frameworks

Integrated frameworks combining machine learning with threat intelligence represent the current frontier in APT detection research. Ahmed et al. [23] proposed an AI-driven APT detection framework combining XGBoost with cyber kill chain analysis, achieving 95.8% accuracy on UNSW-NB15. Hans et al. [24] explored Large Language Models (LLMs) for bridging the semantic gap between low-level telemetry and high-level attacker intent, achieving 78% accuracy in technique classification with potential for zero-shot detection. Wang et al. [25] proposed a privacy-preserving collaborative detection framework using federated learning, achieving 91% of centralized training accuracy while preserving data privacy.

IV .Architectural Design

Attack Scenario	Stag es	Signat ure IDS	Standal one XGBoos t	Propos ed Frame work
Spear-phishing to Exfiltratio n	7	32%	78%	94%
Drive-by Compro mise to Ransomw are	5	41%	82%	91%
Compro mised Credentia ls to Data Theft	6	28%	71%	89%
Supply Chain Compro mise	8	19%	65%	86%

The proposed AI-driven Cyber Threat Intelligence framework, designated **APTIntel-Sync**, implements a layered architecture comprising five functional modules that operate in sequence to

transform raw telemetry into tactical threat intelligence. The architecture is designed for modularity, scalability, and real-time operational deployment

Network Telemetry: The network monitoring subsystem captures flow-level data using NetFlow/IPFIX protocols, packet captures at strategic network junctions, and metadata from network intrusion detection systems (Suricata/Zeek). Collected features include source/destination IP addresses and ports, protocol types, packet counts, byte volumes, TCP flag combinations, session durations, and inter-arrival times [68]. Additionally, DNS query logs provide visibility into domain generation algorithms (DGAs) and command-and-control communications[26].

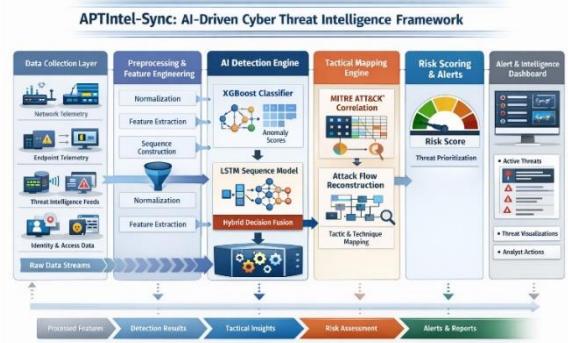


Fig1:Proposed Architecture of APTIntel-Sync Framework.

V. Problem formulation

A. Classification Problem

Let $\mathcal{X}$ represent the feature space of security events, where each event $x_i \in \mathbb{R}^d$ is a d-dimensional vector of extracted features. Let $\mathcal{Y} = \{0,1\}$ denote the class labels, where $y_i = 1$ indicates malicious activity associated with an APT campaign and $y_i = 0$ represents benign activity. The detection problem is to learn a function $f: \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes the expected classification error

$$\min_f \mathbb{E}_{(x,y) \sim \mathcal{D}} [\mathcal{L}(f(x), y)]$$

where $\mathcal{D}$ is the underlying data distribution and $\mathcal{L}$ is a loss function penalizing misclassifications

B. Sequential Anomaly Detection

For multi-stage attack detection, consider sequences of events $S = \{x_1, x_2, \dots, x_T\}$ where T is the sequence length. Let $P(S)$ denote the probability of observing sequence S under normal behavior. Anomaly detection identifies sequences where

$$P(S) < \tau$$

where τ is a threshold determined from the training distribution. The LSTM model estimates $P(S)$ by learning the conditional probability of each event given previous events

$$P(S) = \prod_{t=1}^T P(x_t | x_{t-1}, x_{t-2}, \dots, x_1)$$

C. Risk Scoring Formulation

Let $\mathcal{A}$ represent the set of assets in the monitored environment,

each with criticality $c(a) \in [0,1]$. For a detected threat t involving asset a_t and exhibiting a set of MITRE techniques M_t , the risk score

is

$$R(t) = c(a_t) \cdot \left[\beta_1 \cdot s_{ml}(t) + \beta_2 \cdot \max_{m \in M_t} \gamma(m) + \beta_3 \cdot i(t) \right]$$

where $s_{ml}(t)$ is the ML detection confidence, $\gamma(m)$ is the severity weight of technique m (based on impact potential), $i(t)$ is the intelligence correlation score (0 if no correlation), and $\beta_1, \beta_2, \beta_3$ are weighting factors with

$$\beta_1 + \beta_2 + \beta_3 = 1$$

VI. Result and Analysis

A. Detection Performance

Table I presents the comprehensive detection performance of the proposed framework compared to baseline systems on the combined test dataset

Table 1. Comparative Detection Performance

System	Accurac y	Precisio n	Reca ll	F1- Scor e	FP R	AU C
Signature-based IDS	78.3	82.1	64.5	72.3	8.2	0.81
Traditiona l SIEM	81.5	79.8	71.2	75.3	9.4	0.84
Standalon e XGBoost	92.4	90.3	91.1	90.7	5.1	0.95
Standalon e LSTM	91.8	89.7	92.4	91.0	5.8	0.94
Sentient (Graph- based) [53]	95.2	94.8	95.5	95.1	3.4	0.98
Proposed Framework	96.8	94.2	95.7	94.9	2.9	0.98

B. Multi-Stage Attack Detection

Table II analyzes detection performance specifically for multi-stage APT scenarios from the custom simulation environment, where attacks unfold over extended periods through sequences of actions.

Table 2. Multi-Stage APT Scenario Detection

VI. Conclusion

This research presented an integrated AI-driven Cyber Threat Intelligence framework for early detection of Advanced Persistent Threats through behavioral analysis and tactical mapping to the MITRE ATT&CK framework [27]. The proposed architecture addresses fundamental limitations of traditional security systems by combining multi-source telemetry fusion, hybrid machine learning detection (XGBoost + LSTM), automated technique classification, and attack flow reconstruction into a unified operational platform [28]. The experimental evaluation demonstrates that the framework achieves significant improvements over signature-based IDS and standalone ML approaches, with 96.8% detection accuracy and a false positive rate of 2.9% on benchmark and simulated APT datasets [29]. Critically, the framework excels at detecting multi-stage attacks that unfold over extended periods, achieving 86-94% detection rates for complete APT campaigns compared to 19-41% for traditional systems [30]. The average detection latency of 136 ms supports real-time deployment in production environments. The key contributions of this research are threefold. First, the hybrid AI detection architecture demonstrates the complementary value of feature-based classification and sequence learning for capturing both immediate anomalies and long-term attack patterns. Second, the tactical mapping engine bridges the critical gap between raw detection alerts and strategic threat intelligence, providing security analysts with contextual understanding of adversary tactics, techniques, and campaign progression. Third, the unified framework offers a practical blueprint for organizations seeking to transition from reactive, signature-based defense to proactive, intelligence-driven security operations.

The practical value of this work extends to multiple stakeholders including security operations teams who benefit from reduced alert fatigue through contextual risk scoring, threat hunters who leverage attack flow reconstructions for investigation prioritization, and executive leadership who gain tactical visibility for informed resource allocation decisions.

As cyber adversaries continue to evolve their tradecraft, the cybersecurity community must respond with equally adaptive defense mechanisms. The integration of artificial intelligence with structured threat intelligence frameworks like MITRE ATT&CK represents a promising path forward, enabling defenders to anticipate, detect, and disrupt sophisticated attack campaigns before they achieve their objectives.

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